PATENT COOPERATION TREATY

From the
INTERNATIONAL SEARCHING AUTHORITY

To:
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PCT

WRITTEN OPINION OF THE
INTERNATIONAL SEARCHING AUTHORITY

(PCT Rule 43*bis*.1)

Date of mailing *(day/month/year)*	09-07-2025
Applicant's or agent's file reference 1403122501	**FOR FURTHER ACTION** See paragraph 2 below

International application No.	International filing date *(day/month/year)*	Priority date *(day/month/year)*
PCT/IB2025/052752	16-03-2025	16-03-2025

International Patent Classification (IPC) or both national classification and IPC
A61K38/17,C07K14/705,A61P35/00 Version=2025.01

Applicant
HOMAYOON, SHIDOOSH,MOUSAVIAN HIAGH, ZAHRASADAT, and others.

1. This opinion contains indications relating to the following items:

- [x] Box No. I Basis of the opinion
- [ ] Box No. II Priority
- [ ] Box No. III Non-establishment of opinion with regard to novelty, inventive step and industrial applicability
- [ ] Box No. IV Lack of unity of invention
- [x] Box No. V Reasoned statement under Rule 43*bis*.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement
- [ ] Box No. VI Certain documents cited
- [ ] Box No. VII Certain defects in the international application
- [x] Box No. VIII Certain observations on the international application

2. **FURTHER ACTION**

If a demand for international preliminary examination is made, this opinion will be considered to be a written opinion of the International Preliminary Examining Authority ("IPEA") except that this does not apply where the applicant chooses an Authority other than this one to be the IPEA and the chosen IPEA has notified the International Bureau under Rule 66.1*bis*(b) that written opinions of this International Searching Authority will not be so considered.

If this opinion is, as provided above, considered to be a written opinion of the IPEA, the applicant is invited to submit to the IPEA a written reply together, where appropriate, with amendments, before the expiration of 3 months from the date of mailing of Form PCT/ISA/220 or before the expiration of 22 months from the priority date, whichever expires later.

For further options, see Form PCT/ISA/220.

Name and mailing address of the ISA/	Date of completion of this opinion	Authorized officer
Indian Patent Office Plot No. 32, Sector 14, Dwarka,New Delhi-110075 Facsimile No.	09-07-2025	P Jyothish Kumar Telephone No. +91-1125300200

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY

International application No.
PCT/IB2025/052752

Box No. I Basis of this opinion

1. With regard to the **language**, this opinion has been established on the basis of:

   ☒ the international application in the language in which it was filed.

   ☐ a translation of the international application into _____ which is the language of a translation furnished for the purposes of international search (Rules 12.3(a) and 23.1(b)).

2. ☐ This opinion has been established taking into account the **rectification of an obvious mistake** authorized by or notified to this Authority under Rule 91 (Rule 43*bis*.1(b)).

3. ☐ With regard to any **nucleotide and/or amino acid sequence** disclosed in the international application, this opinion has been established on the basis of a sequence listing:

   a. ☐ forming part of the international application as filed.

   b. ☐ furnished subsequent to the international filing date for the purposes of international search (Rule 13*ter*.1(a)),

      ☐ accompanied by a statement to the effect that the sequence listing does not go beyond the disclosure in the international application as filed.

4. ☐ With regard to any nucleotide and/or amino acid sequence disclosed in the international application, this opinion has been established to the extent that a meaningful opinion could be formed without a WIPO Standard ST.26 compliant sequence listing.

5. Additional comments:

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY

International application No.
PCT/IB2025/052752

Box No. V **Reasoned statement under Rule 43*bis*.1(a)(i) with regard to novelty, inventive step and industrial applicability; citations and explanations supporting such statement**

1. Statement

Novelty (N)	Claims	1-20	YES
	Claims	NONE	NO
Inventive step (IS)	Claims	NONE	YES
	Claims	1-20	NO
Industrial applicability (IA)	Claims	1-20	YES
	Claims	NONE	NO

2. Citations and explanations:

Reference is made to the following document cited in International Search Report:

D1: Sloan, T. L. (2011). Characterization of Receptor Mediated Interactions with Wnt Proteins [Master's thesis, University of California, San Diego].

1. Novelty (Article 33(2) of PCT):

The subject matter of claims 1-20 is novel and meets the requirements of Article 33(2) of PCT as the cited document alone does not disclose the claimed subject matter.

2. Inventive step (Article 33(3) of PCT):

D1 discloses a highly analogous method for producing a recombinant FZD7-CRD fusion protein and confirms its function as a Wnt signaling inhibitor. D1 discloses the understanding of interactions between Wnt proteins and their receptors, particularly the Frizzled (FZD) family. D1 discloses the design of primers, PCR amplification of the FZD7-CRD region, cloning into vectors, and confirmation of protein expression using Western blotting A key part of D1 was the successful cloning, expression, and purification of a recombinant fusion protein, FZD7-CRD-Ig. The CRD is explicitly identified as the domain responsible for binding Wnt ligands. These soluble CRD-Ig fusion proteins can act as inhibitors or antagonists of Wnt signaling. By binding to Wnt proteins in the extracellular space, the FZD7-CRD-Ig protein prevents the Wnt ligands from engaging with their natural cell-surface receptors, thereby blocking the downstream signaling cascade.

WRITTEN OPINION OF THE INTERNATIONAL SEARCHING AUTHORITY

International application No.
PCT/IB2025/052752

Box No. VIII Certain observations on the international application

The following observations on the clarity of the claims, description, and drawings or on the question whether the claims are fully supported by the description, are made:

The phrase "methods for developing the same" in claim 1 encompasses an indeterminate number of possible methods. Claim 1 and its dependent claims dependent claims are merely narrating the invention without clearly mentioning any product features or operative steps (Article 6 of PCT).

Supplemental Box

In case the space in any of the preceding boxes is not sufficient.
Continuation of:

Continuation of Citation and Explanation(Box5)

D1 discloses the significance of FZD7 in the context of cancer and FZD7 is highly expressed in colon cancer cells and is implicated in aberrant Wnt signaling in these cancers. D1 discloses down-regulating FZD7 expression in colon cancer cells leads to decreased survival, invasion, and metastatic capabilities. D1 establishes the methodology for producing the FZD7-CRD protein and confirms its function as a Wnt signaling inhibitor.

The subject matter of claims 1-20 lack an inventive step in view of the disclosure of D1 combined with common general knowledge. D1 discloses a recombinant FZD7-CRD fusion protein and explicitly demonstrates its function as an inhibitor of the Wnt signaling pathway. D1 establishes a clear motivation to use this inhibitor in a cancer context that links FZD7 expression to increased survival, invasion, and metastatic capabilities of colon cancer cells. A person skilled in the art, starting from D1, would be directly led to test the therapeutic potential of the FZD7-CRD inhibitor in relevant cancer models. Investigating the effects of a known pathway inhibitor on well-established hallmarks of cancer, such as tumorigenesis, metastasis, and drug resistance, in a standard model like cancer stem cells, would be a matter of routine experimentation. The subject matter claimed in claims 1-20 is obvious. Therefore, the subject matter of the claims 1-20 is obvious and does not meet the requirements of Article 33(3) of the PCT with regard to inventive step.

3. Industrial Applicability (Article 33(4) of PCT):

Claims 1-20 meet the criteria set out in PCT Article 33(4), and thus have industrial applicability because the subject matter claimed can be made or used in the industry.